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3.3 Spatial Structure: Axons

Any given neuron has a single axon that leaves the soma to make synaptic contacts. Like dendrites, axons have a range of different morphologies. Some axons project mainly to neurons close by. This is the case for neurons in the layer 2-3 of cortex, their axons branch out in all directions from the soma forming a star-shaped axonal arbor called a ‘daisy’. Other neurons such as pyramidal neurons situated deeper in the cortex have axons that plunge in the white matter and may cross the whole brain to reach another brain area. There are even longer axons that leave the central nervous system and travel down the spinal cord to reach muscles at the tip of the foot.

In terms of propagation dynamics, we distinguish two types of axons: the myelinated and the unmyelinated axons. We will see that myelin is useful to increase propagation speed in far-reaching projections. This is the case for cortical projections passing through the white matter, or for axons crossing the spinal cord. Short projections on the other hand use axons devoid of myelin.

3.3.1 Unmyelinated Axons

Mathematical description of the membrane potential in the axon is identical to that of dendrites with active ion channels.

Unmyelinated axons contain sodium and potassium channels uniformly distributed over their entire length. The classical example is the squid giant axon investigated by Hodgkin and Huxley. The Hodgkin-Huxley model described in Chapter 2 was developed for a small axon segment. The general equation for ion channels imbedded on a passive membrane is

$$c r_L \frac{\partial}{\partial t} u(t, x) = \frac{\partial^2}{\partial x^2} u(t, x) - r_L (u(t, x) - E_l) - r_L i_{\text{ion}}[u](t, x). \quad (3.37)$$

where we have reverted to a variable u in units of mV from the equation of active dendrites seen in Sect. 3.2.3. For the giant squid axon, the ionic current are described by the Hodgkin-Huxley model

$$i_{\text{ion}}[u](t, x) = g_{\text{Na}} m^3(t, x) h(t, x) (u(t, x) - E_{\text{Na}}) + g_{\text{K}} n^4(t, x) (u(t, x) - E_{\text{K}}). \quad (3.38)$$

In other systems, the axon may be covered with other types of sodium or potassium ion channels.

When an action potential is fired in the axon initial segment, the elevated membrane potential will depolarize the adjacent axonal segments. Sodium channels farther down the axon which were previously closed will start to open, thereby depolarizing the membrane further. The action potential propagates by activating sodium channels along the cable rather than by spreading the charges such as in a passive dendrite. The properties of the ion channels strongly influence conduction velocity. In the unmyelinated axons of the hippocampus, the conduction velocity axons is 0.25 m/s.

The dynamics described by Eqs. (3.37)–(3.38) reproduce many properties of real axons. In particular, two spikes traveling in opposite direction will collide and annihilate each other. This is unlike wave propagating on water. Another property is reflection at branch points. When the impedance mismatch at the point where a single axon splits into two can be significant, the action potential can reflect and start travelling in the direction it came from.

The solution of Eq. (3.37) with sodium and potassium ion channels such as in Eq. (3.38) can not be written in a closed form. Properties of axonal propagation are either studied numerically (see Sect. 3.4) or with reduced models of ion channels.

Example: Speed of Propagation with Simplified Action Potential Dynamics

For the sake of studying propagation properties, we can replace the active properties of a small axonal segment by a bistable switch (152; 356). We can write the time- and space-dependent membrane potential as

$$c r_L \frac{\partial}{\partial t} u(t, x) = \frac{\partial^2}{\partial x^2} u(t, x) - \frac{r_L g}{1-a} u(t, x) (u(t, x) - 1) (u(t, x) - a). \quad (3.39)$$

where $a < 1/2$ and g are parameters. The membrane potential is scaled such that it rests at zero but may be activated to $u = 1$. The reduced model can switch between $u = 0$ and $u = 1$ if it is pushed above $u = a$, but does not reproduce the full upswing followed by downswing of action potentials.

It turns out that Eq. 3.39 can also be interpreted as a model of flame front propagation. The solution of this equation follows (564)

$$u(x, t) = \frac{1}{1 + \exp\left(\frac{x-vt}{\sqrt{2}\lambda^*}\right)} \quad (3.40)$$

with traveling speed

$$v = \frac{c(1-2a)}{\sqrt{2(1-a)r_L/g}}. \quad (3.41)$$

The propagation velocity depends on the capacitance per unit length c , the longitudinal resistance per unit length r_L , and the excitability parameters g and a .

How does the conduction velocity scale with axon size? Since r_L , c and g themselves depend on the diameter of the axon, we expect the velocity to reflect that relationship. The parameters c and g scale with the circumference of the cellular membrane and therefore scale linearly with the radius ρ . The cytoplasmic resistance per unit length, however, scales with the cross-sectional area, $r_L \propto \rho^2$. With these relations in mind, eq. (3.41) shows that the conduction velocity is proportional to the square root of the diameter $v \propto \sqrt{\rho}$. Therefore, increasing the diameter improves propagation velocity. This is thought to be the reason why the unmyelinated axons that Hodgkin and Huxley studied were so large (up to $\rho = 500 \mu\text{m}$).

3.3.2 Myelinated Axons

Myelinated axons have sodium and potassium channels only in restricted segments called *nodes of Ranvier*. These nodes form only 0.2% of the axonal length, the rest is considered a passive membrane that is wrapped into a myelin sheath. Myelin mainly decreases the membrane capacitance C and increase the resistance R_T by a factor of up to 300 (118). Ions are trapped by myelin since it prevents them to either flow outside the axon or accumulate on the membrane. Instead, ions flow in and out of the nodes such that an ion leaving a node of Ranvier forces another to enter the following node. Assuming that the nodes are equally separated by a myelinated segment of length L , we can model the evolution of the membrane potential at each node u_n . The dynamics of idealized myelinated axons follow Kirchoff's equation with a resistance $R_L = L r_L$ replacing the myelinated segment

$$C \frac{du_n}{dt} = \frac{1}{L r_L} (u_{n+1}(t) - 2u_n(t) + u_{n-1}(t)) - \sum_{ion} I_{ion,n}(t) \quad (3.42)$$

where C is the total capacitance of the node. This equation was encountered in the derivation of the Cable Equation (Section 3.17). The conduction velocity is greatly increased by myelin such that some nerves reach 70-80 m/s (118).

Example: Propagation Speed with Simplified Action Potential Dynamics

Using the simplification of the ion channel dynamics in Eq. (3.39) for each node

$$C \frac{d u_n}{d t} = \frac{1}{L r_L} (u_{n+1}(t) - 2u_n(t) + u_{n-1}(t)) - \frac{g}{1-a} u_n(t) (u_n(t) - 1) (u_n(t) - a) \quad (3.43)$$

where g and $a < 1/2$ are parameters regulating the excitability of the node. Unlike the Eq. (3.39), the parameter g has units of conductance per node since we nodes of Ranvier discrete segments. An activated node may fail to excite the adjacent nodes if the membrane potential does not reach $u = a$. In this model, the internodal distance must satisfy (142)

$$L < L^* = \frac{1-a}{4a^2 r_L g} \quad (3.44)$$

for propagation to be sustained. When the internodal distance L is smaller than L^* , propagation will succeed. When the internodal distance is larger, propagation will fail.

The propagation velocity for small $L^* - L$ follows (59)

$$v \approx \frac{\pi g a}{(1-a) C} \sqrt{L(L^* - L)} \quad (3.45)$$

which is maximum at $L = L^*/2$. Since in most myelinated axons, internodal distance scales linearly with their radius (545), the velocity of myelinated axons also scales linearly with radius, $v \propto L \propto \rho$.